Hedged Straddles, Hedging Error and the Variance Swap

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Abstract

The motivation of the document arises from a natural question: if a continuously delta-hedged straddle generates PnL related to realized variance, why isn't a delta-hedged straddle simply a variance swap? The distinction is subtle but important: the two instruments have different ways of weighting realized variance across spot levels. Understanding this difference is the central theme of this note.

We work through a connected chain of results in volatility trading. First, we prove that the PnL of a continuously delta-hedged long option position equals the integral of dollar gamma against the difference between realized and implied variance, and that this holds for any option, not just a straddle. Second, we isolate vega, vanna and volga terms hidden by marking at constant implied volatility, which distinguishes a variance position held to expiry from a vega position closed early. Third, we examine discrete hedging: the resulting estimator of integrated variance has a hedging error whose standard deviation scales as $1/\sqrt{n}$. Fourth, we compare the hedged straddle with a variance swap and show why the two differ despite both being "bets on variance" — the straddle is gamma-weighted and path-dependent, the swap path-independent — and recover the pure variance payoff through the $1/K^2$-weighted static replication. We close with the two ways the clean picture breaks on a real book: jumps, which separate the replicated log-contract from realized variance and give variance swaps their crash risk, and the gamma swap, the $1/K$-weighted sibling that tames that tail.

Contents

1. Setup and notation

We work under the standard Black–Scholes setup with zero interest rate and zero dividends, both for clarity; reintroducing r and q adds carry terms but does not change the structure of the argument. Let S_t be the underlying price and assume it follows

$$dS_t = \mu_t S_t\,dt + \sigma_t^{\text{real}} S_t\,dW_t, \tag{1}$$

where σ_t^{real} is the (potentially stochastic) realized instantaneous volatility. The trader has bought an option, holds it until expiry T, and marks it on book at a **constant implied volatility** σ^{imp}. Write $V(S,t) = V(S,t;\sigma^{\text{imp}})$ for this marked value. We use the standard Greeks

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}, \qquad \Theta = \frac{\partial V}{\partial t}.$$

By construction V satisfies the Black–Scholes PDE at the implied volatility:

$$\Theta + \tfrac{1}{2}(\sigma^{\text{imp}})^2 S^2 \Gamma = 0. \tag{2}$$

2. The continuous PnL: dollar gamma against variance differential

2.1 Derivation

Apply Itô's lemma to $V(S_t,t)$ under the *realized* dynamics:

$$dV = \Theta\,dt + \Delta\,dS_t + \tfrac{1}{2}\Gamma\,(dS_t)^2. \tag{3}$$

Since $(dS_t)^2 = (\sigma_t^{\text{real}})^2 S_t^2\,dt$, we have

$$dV = \Theta\,dt + \Delta\,dS_t + \tfrac{1}{2}\Gamma(\sigma_t^{\text{real}})^2 S_t^2\,dt. \tag{4}$$

The trader's hedged book is long the option and short Δ units of the underlying. The instantaneous PnL is

$$\begin{aligned} d\Pi &= dV - \Delta\,dS_t \qquad (5)\\ &= \Theta\,dt + \tfrac{1}{2}\Gamma(\sigma_t^{\text{real}})^2 S_t^2\,dt. \qquad (6) \end{aligned}$$

The directional $\Delta\,dS_t$ terms cancel exactly — this is the whole point of delta-hedging. Substituting the BS PDE (2) for Θ,

$$\boxed{d\Pi = \tfrac{1}{2}\Gamma S_t^2\left[(\sigma_t^{\text{real}})^2 - (\sigma^{\text{imp}})^2\right] dt.} \tag{7}$$

Integrating to expiry,

$$\Pi_T = \frac{1}{2}\int_0^T \Gamma_t\,S_t^2\left[(\sigma_t^{\text{real}})^2 - (\sigma^{\text{imp}})^2\right] dt. \tag{8}$$

The quantity $\frac{1}{2}\Gamma_t\,S_t^2$ is therefore central to the interpretation of the trade. It determines how much PnL the option (or the structure of options) generates for a given amount of realized variance. In other words, the delta-hedged option does not simply "own variance"; it owns variance with a weight determined by its dollar gamma.

2.2 Reading the formula

Three observations are worth internalizing.

1. **Variance, not volatility.** The PnL is driven by $(\sigma^{\text{real}})^2 - (\sigma^{\text{imp}})^2$, the difference of *variances*. It is variance, not volatility, that is additive over time.
2. **Direction-free.** Once delta is hedged, the position has no first-order exposure to direction. They are taking a view on whether realized variance will exceed implied variance. Its remaining PnL is driven by the interaction between realized variance and the option's dollar gamma.
3. **Dollar gamma is the weighting.** The quantity ΓS^2, known as *dollar gamma* or *cash gamma*, weights the variance differential. The position only earns variance PnL where gamma exists — near the strike for vanilla options, near expiry for short-dated trades. If S drifts far from the strike, gamma collapses and the position stops harvesting variance even if vol remains elevated.

2.3 Why a straddle? The formula is not special to it

Equation (8) deserves a careful reading, because it was derived for a *generic* marked value $V(S,t)$ and never used any property of a straddle. The result holds verbatim for a single delta-hedged call, a single delta-hedged put, or any European option: the PnL of a continuously delta-hedged long option is always the dollar-gamma-weighted integral of the variance differential. Going long realized variance therefore requires nothing more elaborate than one delta-hedged option. The straddle is not special to the formula.

Why, then, do desks trade the straddle rather than a single option when they want variance exposure? Two reasons, both about the *shape* of the dollar gamma $\Gamma_t S_t^2$ that weights the integral, not about the formula itself.

First, gamma is maximized at the money. Among options of a given maturity, the ATM option carries the most dollar gamma per unit of premium, so it harvests the most variance per euro at risk. This argues for an ATM *strike*, and since the ATM call and the ATM put share the same strike, they carry identical gamma; a straddle simply holds both.

Second, and this is the real motive, the two legs delay the moment at which the position goes silent. A single option loaded at the money still has its gamma concentrated near the strike, so a move in either direction erodes it. The straddle does not remove this decay — it too loses gamma as S leaves the strike — but by centering the maximum-gamma region on the current spot and holding twice the ATM gamma, it keeps the position responsive to variance over a slightly wider band of outcomes before the wings silence it.

> The honest conclusion is that the straddle is a better *single-name* variance harvester than one option, but only marginally: it improves the gamma profile, it does not fix the path-dependence. The instrument that actually removes the path-dependence is not a cleverer option position at all — it is the strip of Section 4.4, whose weighting makes the dollar gamma constant in S. This is the thread that runs through the rest of the note: one option, then the straddle, then the variance swap, each a step closer to pure path-independent variance.

2.4 Vega, vanna, volga: what constant-implied marking hides

The PnL formula (8) was derived by marking the option at a *constant* implied volatility σ^{imp}: the marked value was $V(S,t;\sigma^{\text{imp}})$ with σ^{imp} frozen. Under that assumption V depends on (S, t) only, Itô applies in two variables, and the result is clean. On a real book implied volatility moves day to day, and that motion contributes a term the derivation never saw. Restoring it, the marked value is $V(S, t, \sigma_t^{\text{imp}})$ with σ_t^{imp} now a process. Itô in three variables adds the implied-vol terms to (3),

$$dV = \Theta\, dt + \Delta\, dS_t + \tfrac{1}{2}\Gamma\,(dS_t)^2 + \underbrace{\mathcal{V}\, d\sigma_t^{\text{imp}}}_{\text{vega}} + \underbrace{\frac{\partial \mathcal{V}}{\partial S}\, dS_t\, d\sigma_t^{\text{imp}}}_{\text{vanna}} + \underbrace{\tfrac{1}{2}\frac{\partial \mathcal{V}}{\partial \sigma^{\text{imp}}}\,(d\sigma_t^{\text{imp}})^2}_{\text{volga}},$$

where $\mathcal{V} = \partial V/\partial\sigma^{\text{imp}}$ is vega, $\partial\mathcal{V}/\partial S$ is vanna, and $\partial\mathcal{V}/\partial\sigma^{\text{imp}}$ is volga. Delta hedging still cancels the $\Delta\, dS_t$ term, but it touches none of the implied-vol terms. Using the BS PDE (2) to replace $\Theta\, dt + \frac{1}{2}\Gamma(dS_t)^2$ with the gamma/variance term exactly as before, the hedged PnL is

$$d\Pi = \underbrace{\tfrac{1}{2}\Gamma S_t^2\left[(\sigma_t^{\text{real}})^2 - (\sigma_t^{\text{imp}})^2\right]dt}_{\text{gamma / variance}} + \underbrace{\mathcal{V}\, d\sigma_t^{\text{imp}}}_{\text{vega}} + \underbrace{\frac{\partial \mathcal{V}}{\partial S}\, dS_t\, d\sigma_t^{\text{imp}}}_{\text{vanna}} + \underbrace{\tfrac{1}{2}\frac{\partial \mathcal{V}}{\partial \sigma^{\text{imp}}}\,(d\sigma_t^{\text{imp}})^2}_{\text{volga}}.$$

The first term is the variance bet already analyzed; the remaining three are mark-to-market PnL from the option's own implied volatility moving, present even when realized variance exactly equals implied — even when the first term is zero.

Remark. This is the precise sense in which (8) is the PnL of a *variance* position and not of a *volatility* position. Only the gamma/variance term accumulates; the other three are pure mark-to-market, and they share a single fate. Vega, vanna, and volga are all σ-derivatives of V, and every σ-derivative of the terminal payoff vanishes, since the payoff at T depends on S_T alone (intuitively, an about-to-expire option is just its intrinsic value, with no sensitivity left to implied vol). As $t \to T$ the entire vol-sensitivity block therefore switches off no matter how implied vol moves, and a delta-hedged option *held to expiry* earns the path integral of the variance differential and nothing else. The "pure variance" conclusion is thus not special to vega — it holds for the whole family.

A position *closed early* is the mirror image. None of the three Greeks has vanished — expiry is what killed them — so all three are live in the P&L at the current market marks, and it is worth asking what each one responds to. Model implied vol as its own stochastic process,

$$d\sigma_t^{\text{imp}} = (\cdots)\, dt + \xi_t\, dB_t, \qquad d\langle W, B\rangle = \rho_t\, dt,$$

introducing two new market quantities: the vol-of-vol ξ_t, governing how much implied vol itself moves, and the spot–vol correlation ρ_t between the driver W of spot and the driver B of implied vol. Each of the three terms in $d\Pi$ then loads one of these. Vega multiplies $d\sigma_t^{\text{imp}}$ and so loads the *level* of implied vol. Vanna multiplies the cross term $dS_t\, d\sigma_t^{\text{imp}}$, whose size is set by ρ_t — it is the exposure to spot and vol moving together, which is exactly *skew dynamics*. Volga multiplies $(d\sigma_t^{\text{imp}})^2$, whose size is set by ξ_t — the exposure to *vol-of-vol*, which is *smile convexity*. These last two, vanna and volga, are the skew and smile-convexity PnL a market-maker who unwinds before expiry lives on, and they are the subject of the surface-dynamics discussion.

> This is why the same option is a variance trade to a hold-to-expiry desk and a vega/-vanna/volga trade to a market-maker who will unwind tomorrow, and it is the bridge from this to the surface-dynamics study, where $d\sigma^{\text{imp}}$ — how the whole surface moves with spot and time — is the entire subject.

3. Discrete hedging: the $1/\sqrt{n}$ error

In practice the trader hedges at a finite set of times $0 = t_0 < t_1 < \cdots < t_n = T$. Between hedges, Δ is frozen at $\Delta_k := \Delta(S_{t_k}, t_k)$. We derive the resulting PnL and quantify the error introduced by discretization.

3.1 The discrete PnL formula

A second-order Taylor expansion of V between t_k and t_{k+1} gives

$$\Delta V_k \approx \Theta_k \, \Delta t + \Delta_k (S_{k+1} - S_k) + \tfrac{1}{2}\Gamma_k (S_{k+1} - S_k)^2. \tag{9}$$

The hedge contributes $-\Delta_k(S_{k+1} - S_k)$, so the per-period PnL is

$$\Pi_k \approx \Theta_k \, \Delta t + \tfrac{1}{2}\Gamma_k (S_{k+1} - S_k)^2. \tag{10}$$

Using the BS PDE (2) once more and writing $r_k := (S_{k+1} - S_k)/S_k$ for the return over the kth interval,

$$\Pi_k \approx \tfrac{1}{2}\Gamma_k S_k^2 \left[r_k^2 - (\sigma^{\text{imp}})^2 \Delta t \right]. \tag{11}$$

Summing over all intervals,

$$\Pi_T^{(n)} = \frac{1}{2} \sum_{k=0}^{n-1} \Gamma_k S_k^2 \left[r_k^2 - (\sigma^{\text{imp}})^2 \Delta t \right]. \tag{12}$$

The sum $\sum_k r_k^2$ provides the usual realized-variance estimator based on simple returns. For sufficiently fine sampling, this is asymptotically equivalent to the quadratic variation expressed in log returns. As $n \to \infty$ with $\Delta t \to 0$, equation (12) recovers the continuous result (8).

3.2 Quantifying the hedging error

The realized variance estimator $\sum_k r_k^2$ has the right expectation but nonzero dispersion at finite n. The $1/\sqrt{n}$ convergence of the discrete-hedging error to its continuous limit is the central result of Bertsimas, Kogan and Lo [9]; the computation below recovers the rate in the simplest case: constant (ΓS^2) and constant volatility, so that $r_k \sim \sigma\sqrt{\Delta t}\, Z_k$ with $Z_k \sim N(0, 1)$ i.i.d.

Under these assumptions each $r_k^2 = \sigma^2 \Delta t \, Z_k^2$, and the sum of n independent squared standard normals is by definition a chi-squared variable on n degrees of freedom:

$$\sum_{k=0}^{n-1} r_k^2 = \sigma^2 \Delta t \sum_{k=0}^{n-1} Z_k^2 \overset{d}{=} \sigma^2 \Delta t \cdot \chi_n^2.$$

A χ_n^2 variable has mean n and variance $2n$. The prefactor $\sigma^2 \Delta t$ scales the mean by $\sigma^2 \Delta t$ and the variance by $(\sigma^2 \Delta t)^2$, so, using $\Delta t = T/n$,

$$\mathbb{E}\left[\sum_k r_k^2 \right] = n\, \sigma^2 \Delta t = \sigma^2 T, \qquad \operatorname{Var}\left[\sum_k r_k^2 \right] = 2n\, (\sigma^2 \Delta t)^2 = 2n\, \sigma^4 \frac{T^2}{n^2} = \frac{2\sigma^4 T^2}{n}.$$

The mean is the correct total variance $\sigma^2 T$, confirming the estimator is unbiased; the variance carries the $1/n$ that drives everything below.

To pass from the estimator to the PnL, recall from (12) that the discrete PnL is the estimator scaled by dollar gamma plus a deterministic implied-variance term,

$$\Pi_T^{(n)} = \tfrac{1}{2}\Gamma S^2 \sum_k r_k^2 \; - \; \tfrac{1}{2}\Gamma S^2 (\sigma^{\text{imp}})^2 T.$$

The second term is a constant: it shifts the mean but contributes no variance. Only the first term is random, so with $a = \frac{1}{2}\Gamma S^2$ and $\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$,

$$\mathrm{Var}\big(\Pi_T^{(n)}\big) = \big(\tfrac{1}{2}\Gamma S^2\big)^2 \cdot \frac{2\sigma^4 T^2}{n}, \qquad \mathrm{std}\big(\Pi_T^{(n)}\big) = \tfrac{1}{2}\Gamma S^2 \cdot \frac{\sigma^2 T\sqrt{2}}{\sqrt{n}} = \frac{\Gamma S^2 \sigma^2 T}{\sqrt{2n}}.$$

The standard deviation of the discrete-hedging PnL falls as $1/\sqrt{n}$: doubling the number of hedges reduces the noise by a factor of $\sqrt{2}$.

Remark. The constant-ΓS^2 assumption was made only to isolate the estimator's dispersion; in reality dollar gamma varies along the path, exactly the path-dependence emphasized in Section 4. It can nonetheless be replaced by an effective constant, and the reason is that each term of the PnL sum factorizes into a *slowly-varying weight* and a *fast, mean-zero fluctuation*:

$$\Pi_k \;\propto\; \underbrace{\Gamma_k S_k^2}_{\text{weight}} \cdot \underbrace{\big(r_k^2 - (\sigma^{\text{imp}})^2 \Delta t\big)}_{\epsilon_k,\ \text{mean zero}}.$$

The fluctuations ϵ_k are independent across steps with mean zero, so by the same central-limit argument as above their weighted sum has variance $\sum_k (\Gamma_k S_k^2)^2\, \mathrm{Var}(\epsilon_k)$ — the cross terms vanish by independence. Two things follow. First, the $1/\sqrt{n}$ *scaling* is untouched: it comes from accumulating n independent mean-zero fluctuations, which grow as $\sqrt{n}$ whatever fixed weights multiply them — gamma sets the size of each kick, the central limit theorem sets how kicks of any size accumulate. Second, because dollar gamma drifts smoothly while ϵ_k flips sign every step, the weight enters only through the path-average $\frac{1}{n}\sum_k (\Gamma_k S_k^2)^2$; the single constant $(\Gamma S^2)^2$ of the clean formula is simply replaced by this average, and nothing in the n-counting changes. The practical message is the same scaling read backwards: since the error falls only as $1/\sqrt{n}$, *cutting* it in half requires *quadrupling* the hedges ($n \to 4n$). This $4\times$ rule follows from the scaling alone, so it is robust to path-varying gamma — the leading-order truth on any book.

Remark. This is the essential trade-off behind hedge-frequency selection on a real desk. More frequent hedging tightens the realized-variance estimate, but each hedge crosses the bid–ask, pays exchange fees, and can move the market. The optimum therefore reflects a trade-off between the reduction in hedging error and the transaction costs incurred by each additional hedge. The original treatment is due to Leland [8] and the various refinements of his framework.

4. Hedged straddle vs. variance swap

Both instruments are sometimes described informally as "bets on variance," but they do not provide the same variance exposure. The key distinction is not that one is exposed to variance and the other is not; both are. The distinction is how that variance exposure is weighted across spot levels.

4.1 Different payoffs

4.1.1 Variance swap payoff

A variance swap pays

$$\Pi^{\text{VS}} = N \cdot \left(\sigma_{\text{real}}^2 - K_{\text{var}}^2\right), \tag{13}$$

where σ_{real}^2 is the annualized realized variance over $[0, T]$, N is the variance notional, and K_{var}^2 is the fair variance-swap strike — the level that makes the contract worth zero at inception, so no cash changes hands when it is struck. That fair level is not in general the ATM implied variance: as §(4.3) will show, it is the price of the full $1/K^2$ strip of options, and under skew the heavily weighted out-of-the-money puts push it *above* σ^2_{ATM}.

It equals the level σ^2_{ATM} at which one would mark an ATM straddle only on a flat surface, where every strike carries the same vol — and even then only the subtracted *level* coincides, never the weighting, since the straddle remains gamma-weighted and the swap flat. The worked example of (4.5) lives on such a flat surface, which is why the two reference levels are one number there and would separate under skew. The payoff is linear in realized variance. Unlike a vanilla straddle, the variance exposure does not depend on the spot level at which the returns are realized: only the accumulated squared returns matter.

4.1.2 Hedged straddle payoff

By (8), the hedged straddle pays

$$\Pi^{\text{straddle}} = \frac{1}{2}\int_0^T \Gamma_t S_t^2 \left[\, (\sigma_t^{\text{real}})^2 - (\sigma^{\text{imp}})^2 \,\right] dt. \tag{14}$$

This is path-dependent in two distinct ways. First, the weight $\Gamma_t S_t^2$ is itself spot-shaped: it is largest near the strike and falls away in either wing — and although the S_t^2 factor grows as spot rises, the gamma decay dominates, since $\Gamma_t S_t^2 \propto S_t\,\varphi(d_1)$ and the Gaussian $\varphi(d_1)$ collapses faster than the single surviving power of S_t grows. The weight concentrates around the strike whichever way spot travels.

Second — and this is the point the first does not yet give — that weight acts *pointwise in time*. It is not the *total* realized variance over $[0, T]$ that sets the payoff, but the integral of variance against gamma instant by instant, so *when* each increment of variance arrives, relative to where gamma is at that moment, changes what it contributes. Two paths carrying identical total realized variance can therefore pay very differently — the phenomenon §(4.3) makes concrete.

4.2 A path-dependence example

Consider a long ATM straddle struck at $K = 100$ with $S_0 = 100$ and $\sigma^{\text{imp}} = 20\%$. Suppose two scenarios with identical realized variance over the life:

- *Scenario A:* S oscillates around 100 throughout. Gamma stays high.
- *Scenario B:* S rallies to 130 on day 1 and stays there. Gamma collapses.

A variance swap pays the same amount in both scenarios. The hedged straddle captures most of the variance differential in Scenario A and very little in Scenario B — in B, the gamma weighting silences the rest of the variance that occurs after day 1. The mechanism behind this is the rehedging itself.

Explanation of how to hedge long gamma positions: Delta-hedging a long-gamma position forces a mechanical *buy low, sell high*: when spot rises the straddle's delta grows, so shares must be *sold* to return to neutral; when spot falls the delta shrinks, so shares must be *bought*. The trader does not choose this — it is the automatic consequence of holding the hedge neutral. Each oscillation is one such round-trip, buying below and selling above, and the accumulated round-trips *are* the gamma PnL.

This is why oscillation, not merely magnitude, is what the hedged position harvests. In Scenario A the path crosses the same levels many times, and each crossing is a profitable rehedge; the dollar gamma stays high because spot stays near the strike. In Scenario B the single early move is rehedged once, after which spot sits far from the strike, the dollar gamma has collapsed, and no further round-trips occur no matter how much time passes. The realized variance can be identical; the number of profitable rehedges is not. (The round-trips are not free: the trader pays for them through theta, the time decay that is the standing cost of being long gamma.)

Conclusion: The two paths therefore generate the same realized variance but not the same PnL for the delta-hedged straddle. The reason is not the amount of variance realized, but where that variance occurs relative to the option's strike: the straddle's dollar gamma changes as spot moves.
This is the essential difference between a delta-hedged vanilla option and a pure variance exposure. The next question is how such an exposure can be replicated using vanilla options.

4.3 Static replication of the variance swap

The clean way to convert a hedged option position into a variance swap is to build a portfolio whose dollar gamma is constant in S. The classical replication result, due to Demeterfi–Derman–Kamal–Zou (1999) constructs such a portfolio, up to a forward and cash adjustment. The argument has two pieces. The first is a pathwise identity that rewrites realized variance in terms of the log of the price. Applying Itô to $\log S_t$ under the dynamics $dS_t = \mu_t S_t\,dt + \sigma_t^{\text{real}} S_t\,dW_t$ gives

$$d\log S_t = \Big(\mu_t - \tfrac{1}{2}(\sigma_t^{\text{real}})^2\Big)dt + \sigma_t^{\text{real}}\,dW_t, \qquad \frac{dS_t}{S_t} = \mu_t\,dt + \sigma_t^{\text{real}}\,dW_t.$$

The two share the same drift and diffusion; subtracting twice the first from twice the second cancels both, leaving only the variance:

$$2\,\frac{dS_t}{S_t} - 2\,d\log S_t = (\sigma_t^{\text{real}})^2\,dt.$$

Integrating from 0 to T,

$$\int_0^T (\sigma_t^{\text{real}})^2\,dt = -2\log\left(\frac{S_T}{S_0}\right) + 2\int_0^T \frac{dS_t}{S_t}. \tag{15}$$

This is exact on every path: the drift μ_t has cancelled, so no assumption about it — and no choice of measure — enters.

The identity splits realized variance into two replicable pieces. The right-most integral, $2\int_0^T dS_t/S_t$, is the payoff of holding $1/S_t$ units of the underlying continuously: at each instant one is long $1/S_t$ shares, whose return contributes dS_t/S_t. This is a self-financing dynamic position in the stock alone — the forward hedge — requiring no options. The remaining piece, $-2\log(S_T/S_0)$, is the payoff of a *log contract*: a single European claim depending only on the terminal price S_T. Realized variance is therefore the payoff of a static log contract plus a dynamic stock position, and the entire modelling task reduces to replicating that one log-contract payoff — which is what the Carr–Madan strip below accomplishes.

Because (15) is an identity between *payoffs*, the replication it yields is model-free: it holds path by path, with no expectation and no measure taken anywhere. What *does* require a measure is the *price*. The fair variance strike is the expected payoff,

$$K_{\mathrm{var}}^2 = \mathbb{E}^{\mathbb{Q}}\left[\int_0^T (\sigma_t^{\mathrm{real}})^2\,dt\right],$$

and taking $\mathbb{E}^{\mathbb{Q}}$ reintroduces exactly the drift that cancelled in the identity: under the pricing measure $\mathbb{Q}$ the stock drifts at the risk-free rate, and it is that drift which fixes the forward around which the option strip is priced. So there is no tension between a model-free *payoff* and a measure-dependent *price* — replication is an algebraic fact about the payoff, pricing is a probabilistic fact about its expectation. This is why a variance swap can be quoted directly off the listed option strip without committing to any volatility model: one prices the replicating portfolio, and the portfolio is fixed by algebra alone.

Second, by the Carr–Madan replication identity, any twice-differentiable European payoff $f(S_T)$ can be written as

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+\,dK + \int_F^\infty f''(K)(S_T - K)^+\,dK, \tag{16}$$

where F is the forward. Applying this to $f(S) = -2\log(S/F)$ gives $f'(S) = -2/S$ and $f''(K) = 2/K^2$, so the linear terms are $f(F) + f'(F)(S_T - F) = -\frac{2}{F}(S_T - F)$ and

$$-2\log\left(\frac{S_T}{F}\right) = -\frac{2}{F}(S_T - F) + \int_0^F \frac{2}{K^2}(K - S_T)^+\,dK + \int_F^\infty \frac{2}{K^2}(S_T - K)^+\,dK. \tag{17}$$

Now combine this with the pathwise identity (15), $\int_0^T (\sigma_t^{\mathrm{real}})^2\,dt = -2\log(S_T/S_0) + 2\int_0^T dS_t/S_t$. Substituting (17) for the log term (referenced to F rather than S_0, the two differing only by the constant $-2\log(F/S_0)$) gives

$$\int_0^T (\sigma_t^{\mathrm{real}})^2\,dt = \underbrace{2\int_0^T \frac{dS_t}{S_t} - \frac{2}{F}(S_T - F)}_{\text{dynamic forward hedge}} + \underbrace{\int_0^F \frac{2}{K^2}(K - S_T)^+\,dK + \int_F^\infty \frac{2}{K^2}(S_T - K)^+\,dK}_{\text{static strip of OTM options}} + \text{const.} \tag{18}$$

Read left to right, this is the replication: realized variance equals a self-financing position that holds $1/S_t$ units of the underlying (the forward hedge) plus a *static* book of out-of-the-money options, put below F and calls above, each weighted $2/K^2\,dK$ and held to expiry. The right-hand side is fixed at inception; no rebalancing of the options is ever required, which is what “static” means and what distinguishes this from the dynamic delta-hedge of a single option.

It remains to see *why* this particular weighting is the one that produces a variance swap, and here the dollar gamma is the whole point. A single option struck at K contributes dollar gamma $\frac{1}{2}\Gamma(S;K)\,S^2$, a bump peaked near $S = K$. Weighting each strike by $2/K^2$ and integrating,

$$\frac{1}{2}S^2 \int_0^\infty \frac{2}{K^2}\,\Gamma(S;K)\,dK = \text{const}, \qquad \text{independent of } S, \tag{19}$$

because the $1/K^2$ factor is exactly the weight under which the individual gamma bumps, each localized at its own strike, sum to a flat profile across S. The strip's dollar gamma is therefore constant in spot. Contrast this with a single option or a straddle, whose dollar gamma $\frac{1}{2}\Gamma_t S_t^2$ collapses as S leaves the strike — the exact path-dependence of (8) and of the worked example in §4.5. Feeding a constant dollar gamma back into the hedged-option PnL (8),

$$\Pi_T = \frac{1}{2}\int_0^T \Gamma_t S_t^2 \left[(\sigma_t^{\mathrm{real}})^2 - (\sigma^{\mathrm{imp}})^2\right] dt \;\longrightarrow\; \frac{c}{2}\int_0^T \left[(\sigma_t^{\mathrm{real}})^2 - (\sigma^{\mathrm{imp}})^2\right] dt, \tag{20}$$

the gamma weight comes out of the integral and the position pays the *unweighted* integrated variance differential — level-independent, path-independent, exactly the variance-swap payoff of (13). The $1/K^2$ strip is thus the precise device that removes the dollar-gamma weighting the single option could not shed.

4.4 CBOE VIX methodology

The CBOE VIX methodology is a discrete implementation of this model-free variance-replication framework using SPX options over a 30-day horizon. Its defining formula

$$\mathrm{VIX}^2 \;=\; \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rT} Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2$$

is the discrete counterpart of the integrals derived above, and the reader should recognize each piece rather than take it on faith. The $1/K_i^2$ weight on each option is exactly the $f''(K) = 2/K^2$ of the log-contract; the sum over strikes is the discretized strip integral; the ΔK_i are the strike spacings that approximate dK; the factor e^{rT} returns the option mid-prices $Q(K_i)$ to forward value; and the final $(F/K_0 - 1)^2$ term is the correction that arises because the discrete strip is anchored at the first strike K_0 below the forward rather than exactly at F. The $2/T$ prefactor annualizes.

The one genuinely new ingredient in the exchange formula is not conceptual but practical: which options enter the sum. CBOE includes only out-of-the-money quotes — puts below K_0, calls above — and truncates the wings where quotes become unreliable, precisely the liquidity boundary discussed for skew estimation elsewhere in this series. The mathematics is the replication above; the engineering is the truncation rule. For the exact conventions the reader should consult the CBOE VIX white paper, since the index is a *definition* with fixed choices, not a derivation with free ones.

4.5 A worked example: straddle vs. variance swap in numbers

To make the path-dependence concrete, take an ATM straddle on a one-month horizon with $S_0 = K = 100$, $\sigma^{\mathrm{imp}} = 20\%$, $T = \frac{1}{12}$. Under Black–Scholes with zero rates each leg costs about 2.30, so the straddle costs about 4.60. For the gamma, the ATM approximation $\Gamma \approx \varphi(0)/(S\sigma\sqrt{T})$ with $\varphi(0) = 1/\sqrt{2\pi} \approx 0.399$ gives, for a single leg,

$$\Gamma \approx \frac{0.399}{100 \cdot 0.20 \cdot \sqrt{1/12}} = \frac{0.399}{5.77} \approx 0.069;$$

the straddle carries twice this, $\Gamma_{\mathrm{str}} \approx 0.138$, so its dollar gamma at inception is $\frac{1}{2}\Gamma_{\mathrm{str}} S^2 \approx \frac{1}{2}(0.138)(100^2) \approx 691$. The daily breakeven move is $\sigma^{\mathrm{imp}}/\sqrt{252} \approx 1.26\%$, the rule of 16.

Now consider two price paths over the 20 trading days, constructed to have identical realized variance — each day moves the log-price by the same magnitude $|\Delta \log S| = 0.017$, so both realize an annualized volatility of about 26.3%, comfortably above the 20% implied. The paths differ only in the sign pattern of the daily moves:

- **Path A (oscillation):** the sign alternates every day, so spot bounces between 100 and 101.7 and ends back at the strike. Dollar gamma stays high — and, with spot pinned near the strike, sharpens toward expiry.
- **Path B (trend):** the sign is constant, so spot drifts steadily upward and ends near 140. Dollar gamma decays as spot leaves the strike.

Tracking the weight $\frac{1}{2}\Gamma_t S_t^2$ along each path makes the difference concrete:

	Path A (oscillation)		Path B (trend)	
Day	spot	$\frac{1}{2}\Gamma_t S_t^2$	spot	$\frac{1}{2}\Gamma_t S_t^2$
0	≈ 100	691	100.0	691
4	≈ 100	772	107.0	336
8	≈ 100	892	114.6	9
12	≈ 100	1092	122.6	≈ 0
16	≈ 100	1545	131.3	≈ 0

The snapshots fall on even days, where Path A sits at 100; between them it swings to 101.7 and back, realizing its variance through daily ±1.7% moves *while pinned to the strike*. Its dollar gamma therefore stays high and rises toward expiry, as the ATM gamma sharpens when $T - t \to 0$. Path B's weight is already an order of magnitude lower by day 4 and effectively dead by day 8: the variance it realizes thereafter lands on a gamma that has collapsed. Since the straddle harvests variance in proportion to this weight (8), Path A should capture almost all of its realized variance and Path B very little — which is what the simulated PnL confirms:

	Realized vol	Straddle PnL	Variance-swap PnL
Path A (oscillation)	26.3%	+2.00	$+0.0294 \times N$
Path B (trend)	26.3%	+0.44	$+0.0294 \times N$

The variance swap pays identically on both paths because it sees only total realized variance, equal by construction; its payoff is $N(\sigma^2_{\text{real}} - K^2_{\text{var}}) = N(0.0694 - 0.0400) = 0.0294\,N$. Two inputs set that number. The realized side, $\sigma^2_{\text{real}} = 0.0694$, is the annualized realized variance of either path ($\sqrt{0.0694} \approx 26.3\%$). The strike, $K^2_{\text{var}} = 0.0400 = 0.20^2$, equals the implied variance *here* only because the example lives on a flat Black–Scholes surface: with 20% vol at every strike, the $1/K^2$ strip that prices the swap (4.3) is built entirely from 20%-vol options, so its price — the fair variance strike — is exactly 0.20^2. The wedge between K^2_{var} and the ATM quote that §4.8 stresses is a skew effect, absent here by construction; on a real, skewed surface the two would differ and one would read K^2_{var} off the strip.

The straddle does not pay identically: +2.00 on A against +0.44 on B, roughly four to one, despite identical realized variance. By (8) it weights each instant of variance by $\Gamma_t S_t^2$, and the table above shows that weight staying large on A and collapsing on B. Same variance, same variance-swap payoff, very different straddle PnL.

These numbers are the concrete face of the construction in §4.3. An instrument that pays on realized variance *regardless of where it occurs* needs a position whose dollar gamma does not depend on spot — exactly what the $1/K^2$ strip delivers, replacing the straddle's spot-dependent weight with a gamma flat in S, so that identical realized variance always pays identically.

4.6 When replication fails: jumps

Every result so far rests on the diffusion assumption (1): S moves continuously, so $(dS_t)^2 = (\sigma_t^{\text{real}})^2 S_t^2\, dt$ and Itô applies cleanly. Real prices jump — on earnings, on macro releases, on gaps — and a jump breaks the derivation at its root. Understanding exactly *where* it breaks is what separates a variance swap that is priced correctly from one that blows up in a crash.

The static replication of Section (4.3) is exact: the $1/K^2$ strip reproduces the log-contract $-2\log(S_T/F)$ on *every* terminal value S_T, jump or no jump. What is *not* exact once jumps are present is the identity (15) linking the log-contract to realized variance. That identity used Itô, which assumes continuity. When S jumps by a multiplicative factor $1 + J$ (so $S \to S(1 + J)$), the two sides diverge.

Concretely, a single jump of size J contributes $\log^2(1 + J)$ to realized (log-return) variance, but contributes $2\big[J - \log(1 + J)\big]$ to the strip-replicated payoff. Expanding both in J,

$$2\big[J - \log(1 + J)\big] = J^2 - \tfrac{2}{3}J^3 + \cdots\,, \qquad \log^2(1 + J) = J^2 - J^3 + \cdots\,,$$

so the two agree to second order and differ at $O(J^3)$. For small moves the gap is negligible — this is why the replication is excellent in calm markets. For a large move it is not, and the asymmetry matters: a large *downward* jump ($J < 0$) makes $-\log(1 + J)$ grow without bound as $1 + J \to 0$, so the strip payoff *exceeds* the realized variance it was meant to match. The replicating portfolio over-delivers on crashes relative to a contract written on squared returns.

Remark. This is not a rounding error; it is the structural reason variance swaps carry crash risk and why the market moved to *capped* variance swaps and *gamma swaps* after 2008. A short variance-swap position — the natural carry trade, collecting the variance premium — is implicitly short the $O(J^3)$ jump term, which is exactly the left tail. The clean gamma-against-variance picture of (8) is the leading-order truth; the jump residual is the tail that the leading order cannot see, and it is where short variance positions are actually killed.

The honest one-line summary: the strip replicates the payoff exactly, but "the payoff" is the log-contract, and the log-contract is realized variance only when the path is continuous.

4.7 A one-parameter family: the gamma swap

The variance swap was engineered by choosing the option weight $1/K^2$ so that the portfolio's dollar gamma is constant in S. That weight is a choice, not a necessity, and varying it produces a family of variance-like instruments. The most important sibling is the *gamma swap*, obtained by weighting the strip as $1/K$ instead of $1/K^2$.

The consequence is immediate from the same machinery. Where the variance swap pays $\int_0^T (\sigma_t^{\text{real}})^2\, dt$ — each instant of variance weighted equally — the gamma swap pays

$$\int_0^T \frac{S_t}{S_0}\, (\sigma_t^{\text{real}})^2\, dt,$$

each instant of variance weighted by the spot level at that instant. The $1/K$ weighting makes the portfolio's dollar gamma grow linearly with S rather than staying flat, so the instrument accumulates more variance exposure when the underlying is high and less when it is low.

Two properties follow, and both are why the gamma swap exists. First, it is *self-weighting away from crashes*: because low-S variance is down-weighted, a gamma swap is far less exposed to the large downward jumps that damage a plain variance swap — the very $O(J^3)$ tail of 4.6 is suppressed where it hurts most. Second, its replication needs no $1/K^2$-driven emphasis on deep out-of-the-money puts, so it depends less on the illiquid far wing and is easier to hedge in practice.

Remark. The pair is worth holding in mind as two points on a continuum. The variance swap ($1/K^2$) delivers pure, level-independent variance at the cost of heavy wing dependence and crash sensitivity; the gamma swap ($1/K$) trades a little purity for a tamer tail and easier replication. Which one a desk prefers is a statement about how it wants to be exposed to the left tail, not about which is "correct." The weighting is the free parameter, and the choice of weighting *is* the trade.

4.8 Pricing in practice: why the swap never leaves the strip

A variance swap is cash-settled. At expiry one party pays $N\left(\sigma^2_{\text{real}} - K^2_{\text{var}}\right)$, the other receives it, and no option changes hands between them. It is tempting to conclude that the replication of 4.3 is a way to *understand* the payoff but not something the trade depends on — that one could simply agree the payoff and never touch a strip. This subsection argues the opposite. Replication is not scaffolding around the swap; it sets the only free number in the contract, and every practical difficulty of trading one is a difficulty of that replication in a different costume.

The strike is the replication. The payoff fixes exactly one number at inception: the strike K^2_{var} — the quantity one is tempted to call "$\sigma^{\text{imp}\,2}$." Everything else is observed after the fact. So the entire economic content of entering the swap is the choice of that strike, and the strike must be *fair*: chosen so the contract is worth approximately zero at inception, with neither side handed free money on day one. A fair strike is, by definition, the risk-neutral expectation of what the swap will pay,

$$K^2_{\text{var}} = \mathbb{E}^{\mathbb{Q}}\left[\int_0^T (\sigma_t^{\text{real}})^2\,dt\right]. \tag{21}$$

But this expectation is not observable — it is a forward-looking statement about variance that has not yet happened. The only market instrument that pins it down is the option strip: by the replication identity of §4.3, the expectation (21) *equals* the $1/K^2$-weighted price of the out-of-the-money strip. The strike of a variance swap is read off the strip. That is where "σ^{imp}" comes from; it is not a number the two counterparties invent. Replication is already inside the payoff — it is the K^2_{var} one was treating as given.

Remark (The no-arbitrage anchor)**.** Even a purely cash-settled swap, with no options in its mechanics, is chained to the strip through its price. Suppose the strip prices fair variance at level K^2_{strip} but a dealer quotes a swap struck cheaper, $K^2_{\text{deal}} < K^2_{\text{strip}}$. Go long the swap and short the replicating strip. The swap pays $\int\sigma^2 - K^2_{\text{deal}}$; the short strip pays $K^2_{\text{strip}} - \int\sigma^2$; the realized variance cancels between the two legs and $K^2_{\text{strip}} - K^2_{\text{deal}} > 0$ is locked in risk-free (scaling and discounting suppressed). Any strike other than the strip's fair value is an arbitrage, so the traded strike must equal it. The replicating portfolio is the no-arbitrage anchor whether or not a single unit of it is ever held.

This is the mirror image of the payoff-versus-price remark in §4.3. There, the payoff *identity* was model-free algebra while the *price* required a measure. Here the payoff $N(\sigma^2_{\text{real}} - \sigma^2_{\text{imp}})$ looks free of replication precisely because one is staring at the payoff — but the "σ^{imp}" inside it *is* the price, and the price is the strip.

Trading OTC does not escape replication — it relocates it. One can, of course, call a dealer and receive a clean payoff on one's own books with no strip in sight. But the dealer prices and hedges that payoff *with* the strip; the quote returned is K^2_{strip} plus a margin for the frictions catalogued below. Replication does not vanish when the trade goes OTC — it moves from the client to the dealer, who prices it straight back into the quote. The consequence is visible in the market. Where replication is easy — index options, SPX above all — swaps quote tight and trade in size: the market is deep *because* the underlying is cheap to replicate. Where replication is hard — thin single names, unquotable wings, a jumpy underlier — the dealer quotes a wide, defensive level or declines to make a two-way market. The illiquidity one would have hit building the strip oneself does not disappear; it reappears as the dealer's spread. "Trade it OTC" and "worry about replication" are not alternatives: the OTC price *is* the replication cost, marked up.

Remark. One genuine exception. A dealer whose existing client flow already leaves them long the variance a client wants to sell can internalise the trade by netting against that book rather than hedging from scratch, and may quote inside pure replication cost. That is a franchise-and-flow advantage, not an escape from replication economics — it is replication against an existing book instead of against the market.

How hard is it, really? The cost stack. The recipe — buy the strip, delta-hedge the forward — is operationally routine and, on SPX, fully industrialised. The difficulty lives entirely where reality departs from the continuous, fully-strikeable, frictionless idealisation, and those departures are the ones §4.3 and §4.6 already named. Ordered by how much they hurt, and by whether capital and care can remove them:

1. **The un-buyable wing (structural, unhedgeable).** The $1/K^2$ weight loads the largest quantities onto the lowest-strike puts — exactly the strikes that stop being quoted. Past the last live strike one cannot buy the required quantity at any price, so every real book truncates and thereby *under*-replicates the left tail, the region that matters most in a crash. This is not slippage; it is an impossibility past a point, and no amount of capital removes it.

2. **The jump residual (structural, unhedgeable).** Even a perfectly executed, fully-strikeable, continuously-rebalanced strip replicates the *log-contract*, and the log-contract is realized variance only in the continuous limit. A jump separates them at $O(J^3)$ §4.6, asymmetrically — the downside gap over-delivers. No denser strip and no extra diligence closes this gap; it is a divergence between two different payoffs, and it lives in the tail.

3. **Slippage, entry/exit and running (large on thin names, small on SPX).** Dozens of legs, each crossing a bid–ask twice; plus the *running* cost of the forward delta-hedge, every rehedge crossing the spread — the transaction-cost side of the §3 hedging-error trade-off. This is the honest content of the intuition that "the contracts and slippage would kill it," and it is correct — for single names. On SPX it is small enough to absorb.

4. **Re-striking turnover.** The "static" strip is static only in the idealised picture. As spot drifts and options roll off, the moneyness grid shifts and the book must be re-struck,

converting part of the "static" position into further turnover.

5. **Discretisation basis (minor, shrinkable).** A finite strike grid (ΔK_i) and discrete rehedging ($1/\sqrt{n}$, 3.2): accepted errors, tightened by a denser grid and a larger notional, never a threat.

The top two are structural — no capital or care removes them; the bottom three are frictions — large on thin names, negligible on SPX. This is exactly why single-name variance is an OTC dealer product while index variance is industrialised, and it is the precise sense in which "slippage kills this" is right where it is right.

Why the cap goes on the left tail. After 2008 the market moved to *capped* variance swaps, which bound the payoff. The cap sits on the left tail, not the right, and the asymmetry is not a convention: three independent reasons all point the same way.

- **The replication error is itself one-sided.** A downward jump sends $-\log(1+J) \to \infty$ as $1+J \to 0$ §4.6: the strip over-delivers on crashes without bound. The upside carries no analogous blow-up. The term hardest to replicate and most explosive is specifically the left one.
- **Equities crash down and grind up.** Realized variance is driven by the large gap, and for equities the large gap is overwhelmingly a downside gap — the same left-skew the event notes assume. Melt-ups exist but are rarer and smaller; the left tail is where the notional is actually hit.
- **The short-variance seller is the one who dies.** The natural trade is *selling* variance to harvest the variance risk premium; the seller is short the left tail, and uncapped, a single crash can exceed years of premium — the negative-skew arithmetic of the naked event trade. The cap is what makes the short survivable, hence what makes the seller willing to quote at all. Nobody needs protection from a melt-up.

All three arrows point left: the left tail is simultaneously the hardest to replicate, the most likely to be realized, and the one that bankrupts the short. The right tail is none of these.

The market's verdict. The cleanest answer to "how hard is replication" is what the market built in response to it. It moved to *capped* variance swaps — bounding the un-replicable left tail directly — and toward the *gamma swap* of §4.7, whose $1/K$ weighting down-weights low-S variance and so leans far less on both the illiquid deep-put wing and the $O(J^3)$ tail. Both are engineered admissions that the pure $1/K^2$ variance swap is not cleanly replicable in the tail; each re-tools the contract around the *top* of the cost stack rather than the bottom.

Remark (The single sentence to keep)**.** The bulk of a variance swap is cheap and industrialised to replicate; its left tail is impossible to replicate in practice — which is why the strike is priced off the strip, why single-name swaps live OTC, why the cap goes on the left, and why the pure variance swap partly gave way to its capped and gamma-weighted siblings. Nothing about the swap ever leaves the strip; the strip only becomes harder to hold as one moves into the tail.

5. Conclusions

A few takeaways worth keeping at hand.

1. A continuously delta-hedged long option earns $\frac{1}{2}\int \Gamma S^2[(\sigma^{\text{real}})^2 - (\sigma^{\text{imp}})^2]\,dt$. The first-order directional exposure is hedged, leaving a variance exposure weighted by dollar gamma. The formula is not special to the straddle: it holds for any single option (2.3).

2. Marking at constant implied vol hides vega, vanna and volga (2.4). Held to expiry the whole vol-sensitivity block vanishes and the option is a pure variance trade; closed early it is realized in full and becomes a vega/vanna/volga trade, loading level, spot–vol correlation and vol-of-vol. This is the bridge to the surface-dynamics notes.

3. Under the constant-volatility and constant-gamma approximation considered here, the standard deviation of the discretization error scales as $1/\sqrt{n}$. Doubling hedge frequency cuts the noise by $\sqrt{2}$, so *halving* it costs four times the hedges — a scaling robust to path-varying gamma, and paid for in transaction costs.

4. The hedged straddle is gamma-weighted and path-dependent; two paths with the same realized variance can produce very different PnL. The variance swap removes the gamma weighting through the $1/K^2$ strip, which is also the foundation of the VIX computation (4.4).

5. The clean picture breaks in two identifiable ways on a real book. Jumps separate the replicated log-contract from realized variance at $O(J^3)$, giving variance swaps their crash risk (4.6); and the weighting is a free parameter, with the gamma swap ($1/K$) trading purity for a tamer tail (4.7). The variance/gamma decomposition remains the right leading-order starting point — the residuals are where a real book lives.

6. A variance swap never leaves the strip (4.8). Its strike is not an input the counterparties invent but the price of the $1/K^2$ strip, so "σ^{imp}" in the payoff *is* the replication; trading OTC relocates that replication to the dealer rather than escaping it. What is cheap to replicate (the bulk, on liquid index surfaces) quotes tight; what is impossible (the un-buyable deep-put wing and the $O(J^3)$ jump residual) is why single-name swaps live OTC, why the cap sits on the left tail, and why the pure variance swap gave ground to its capped and gamma-weighted siblings.

> The result should not be interpreted as saying that a delta-hedged straddle is simply a poor-man's variance swap. The two positions have different state-dependent exposures. A straddle concentrates variance exposure around its strike, whereas the variance replication deliberately constructs an approximately uniform dollar-gamma exposure across spot levels.

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